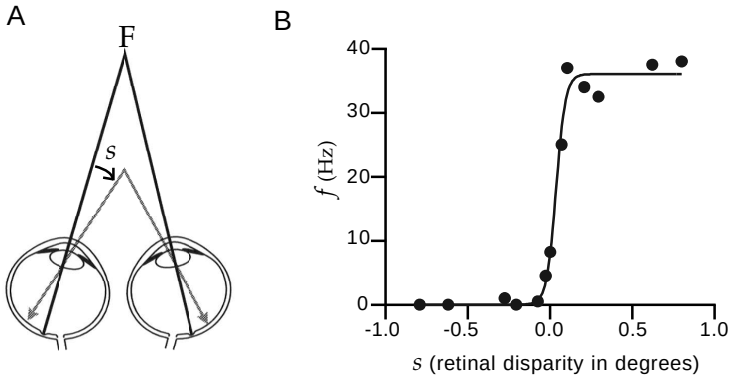

Figure 1.7 (A) Definition of retinal disparity. The gray lines with arrows show the location on each retina of an object located nearer than the fixation point F. The image from the fixation point falls at the fovea in each eye, the small pit where the black lines meet the retina. The image from a nearer object falls to the left of the fovea in the left eye and to the right of the fovea in the right eye. For objects farther away than the fixation point, this would be reversed. The disparity angle s is indicated in the figure. (B) Average firing rate of a cat V1 neuron responding to separate bars of light illuminating each eye, plotted as a function of the disparity. Because this neuron fires for positive s values, it is called a far-tuned cell. The curve is a fit using the function 1.17 with parameters $r_{\max} = 36.03$ Hz, $s_{1/2} = 0.036^\circ$, and $\Delta_s = 0.029^\circ$. (A adapted from Wandell, 1995; B data points from Poggio and Talbot, 1981.)

$\langle r \rangle = f(s)$ from trial to trial. While the map from stimulus to average response may be described deterministically, it is likely that single-trial responses such as spike-count rates can be modeled only in a probabilistic manner. For example, r values can be generated from a probability distribution with mean $f(s)$. The trial-to-trial deviation of r from $f(s)$ is considered to be noise, and such models are often called noise models. The standard deviation for the noise distribution either can be independent of $f(s)$, in which case the variability is called additive noise, or it can depend on $f(s)$. Multiplicative noise corresponds to having the standard deviation proportional to $f(s)$.

Response variability extends beyond the level of spike counts to the entire temporal pattern of action potentials. Later in this chapter, we discuss a model of the neuronal response that uses a stochastic spike generator to produce response variability. This approach takes a deterministic estimate of the firing rate, $\mathrm{r}_{\mathrm{est}}(t)$, and produces a stochastic spiking pattern from it. The spike generator produces variable numbers and patterns of action potentials, even if the same estimated firing rate is used on each trial.

1.3 What Makes a Neuron Fire?

Response tuning curves characterize the average response of a neuron to a given stimulus. We now consider the complementary procedure of av-

eraging the stimuli that produce a given response. To average stimuli in this way, we need to specify what fixed response we will use to "trigger" the average. The most obvious choice is the firing of an action potential. Thus, we ask, "What, on average, did the stimulus do before an action potential was fired?" The resulting quantity, called the spike-triggered average stimulus, provides a useful way of characterizing neuronal selectivity. Spike-triggered averages are computed using stimuli characterized by a parameter $s(t)$ that varies over time. Before beginning our discussion of spike triggering, we describe some features of such stimuli.

Describing the Stimulus

Neurons responding to sensory stimuli face the difficult task of encoding parameters that can vary over an enormous dynamic range. For example, photoreceptors in the retina can respond to single photons or can operate in bright light with an influx of millions of photons per second. To deal with such wide-ranging stimuli, sensory neurons often respond most strongly to rapid changes in stimulus properties and are relatively insensitive to steady-state levels. Steady-state responses are highly compressed functions of stimulus intensity, typically with logarithmic or weak power-law dependences. This compression has an interesting psychophysical correlate. Weber measured how different the intensity of two stimuli had to be for them to be reliably discriminated, the "just noticeable" difference Δs. He found that, for a given stimulus, Δs is proportional to the magnitude of the stimulus s, so that $\Delta s/s$ is constant. This relationship is called Weber's law. Fechner suggested that noticeable differences set the scale for perceived stimulus intensities. Integrating Weber's law, this means that the perceived intensity of a stimulus of absolute intensity s varies as $\log s$. This is known as Fechner's law.

Weber's law

Fechner's law

Sensory systems make numerous adaptations, using a variety of mechanisms, to adjust to the average level of stimulus intensity. When a stimulus generates such adaptation, the relationship between stimulus and response is often studied in a potentially simpler regime by describing responses to fluctuations about a mean stimulus level. In this case, $s(t)$ is defined so that its time average over the duration of a trial is 0, $\int_0^T dt\, s(t)/T = 0$. We frequently impose this condition.

$\int_0^T dt\, s(t)/T = 0$

Our analysis of neural encoding involves two different types of averages: averages over repeated trials that employ the same stimulus, which we denote by angle brackets, and averages over different stimuli. We could introduce a second notation for averages over stimuli, but this can be avoided when using time-dependent stimuli. Instead of presenting a number of different stimuli and averaging over them, we can string together all of the stimuli we wish to consider into a single time-dependent stimulus sequence and average over time. Thus, stimulus averages are replaced by time averages.

stimulus and time averages

Although a response recorded over a trial depends only on the values

taken by $s(t)$ during that trial, some of the mathematical analyses presented in this chapter and in chapter 2 are simplified if we define the stimulus at other times as well. It is convenient if integrals involving the stimulus are time-translationally invariant so that for any function h and time interval τ

$$\int_0^T dt\, h(s(t+\tau)) = \int_\tau^{T+\tau} dt\, h(s(t)) = \int_0^T dt\, h(s(t))\,. \tag{1.18}$$

To assure the last equality, we define the stimulus outside the time limits of the trial by the relation $s(T+\tau) = s(\tau)$ for any τ, thereby making the stimulus periodic.

periodic stimulus

The Spike-Triggered Average

The spike-triggered average stimulus, $C(\tau)$, is the average value of the stimulus a time interval τ before a spike is fired. In other words, for a spike occurring at time t_i, we determine $s(t_i - \tau)$, and then we sum over all n spikes in a trial, $i = 1, 2, \ldots, n$, and divide the total by n. In addition, we average over trials. Thus,

spike-triggered average $C(\tau)$

$$C(\tau) = \left\langle \frac{1}{n}\sum_{i=1}^{n} s(t_i - \tau) \right\rangle \approx \frac{1}{\langle n\rangle}\left\langle \sum_{i=1}^{n} s(t_i - \tau) \right\rangle . \tag{1.19}$$

The approximate equality of the last expression follows from the fact that if n is large, the total number of spikes on each trial is well approximated by the average number of spikes per trial, $n \approx \langle n\rangle$. We make use of this approximation because it allows us to relate the spike-triggered average to other quantities commonly used to characterize the relationship between stimulus and response (see below). Figure 1.8 provides a schematic description of the computation of the spike-triggered average. Each time a spike appears, the stimulus in a time window preceding the spike is recorded. Although the range of τ values in equation 1.19 is unlimited, the response is typically affected only by the stimulus in a window a few hundred milliseconds wide immediately preceding a spike. More precisely, we expect $C(\tau)$ to approach 0 for positive τ values larger than the correlation time between the stimulus and the response. If the stimulus has no temporal correlations with itself, we also expect $C(\tau)$ to be 0 for $\tau < 0$, because the response of a neuron cannot depend on future stimuli. In practice, the stimulus is recorded only over a finite time period, as indicated by the shaded areas in figure 1.8. The recorded stimuli for all spikes are then summed and the procedure is repeated over multiple trials.

The spike-triggered average stimulus can be expressed as an integral of the stimulus times the neural response function of equation 1.1. If we replace the sum over spikes with an integral, as in equation 1.2, and use the approximate expression for $C(\tau)$ in equation 1.19, we find

$$C(\tau) = \frac{1}{\langle n\rangle}\int_0^T dt\, \langle \rho(t)\rangle\, s(t-\tau) = \frac{1}{\langle n\rangle}\int_0^T dt\, r(t) s(t-\tau)\,. \tag{1.20}$$

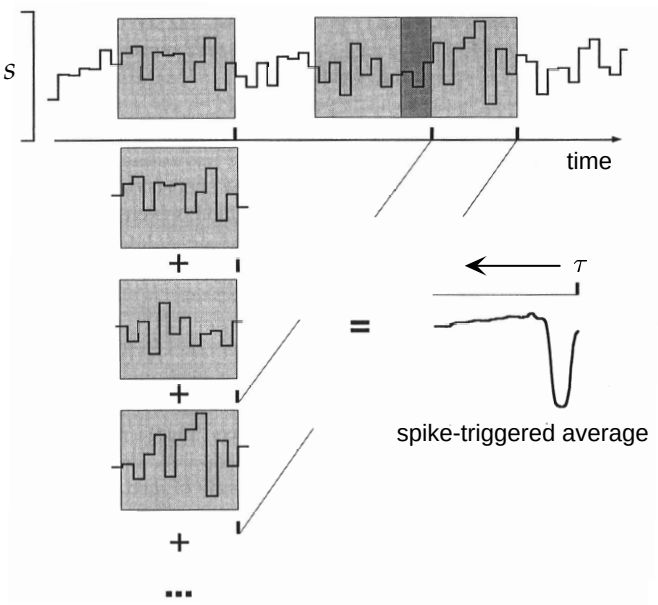

Figure 1.8 Schematic of the procedure for computing the spike-triggered average stimulus. Each gray rectangle contains the stimulus prior to one of the spikes shown along the time axis. These are averaged to produce the waveform shown at the lower right, which is the average stimulus before a spike. The stimulus in this example is a piecewise constant function of time. (Adapted from Rieke et al., 1997.)

The second equality is due to the equivalence of $\langle \rho(t) \rangle$ and $\mathsf{r}(t)$ within integrals. Equation 1.20 allows us to relate the spike-triggered average to the correlation function of the firing rate and the stimulus.

firing-rate stimulus correlation function Q_{rs}

Correlation functions are a useful way of determining how two quantities that vary over time are related to one another. The two quantities being related are evaluated at different times, one at time t and the other at time $t + \tau$. The correlation function is then obtained by averaging their product over all t values, and it is a function of τ. The correlation function of the firing rate and the stimulus is

$$Q_{rs}(\tau) = \frac{1}{T} \int_0^T dt\, \mathsf{r}(t) s(t + \tau) \,. \tag{1.21}$$

By comparing equations 1.20 and 1.21, we find that

$$C(\tau) = \frac{1}{\langle r \rangle} Q_{rs}(-\tau) \,, \tag{1.22}$$

reverse correlation function

where $\langle r \rangle = \langle n \rangle / T$ is the average firing rate over the set of trials. Because the argument of the correlation function in equation 1.22 is $-\tau$, the spike-triggered average stimulus is often called the reverse correlation function. It is proportional to the correlation of the firing rate with the stimulus at preceding times.

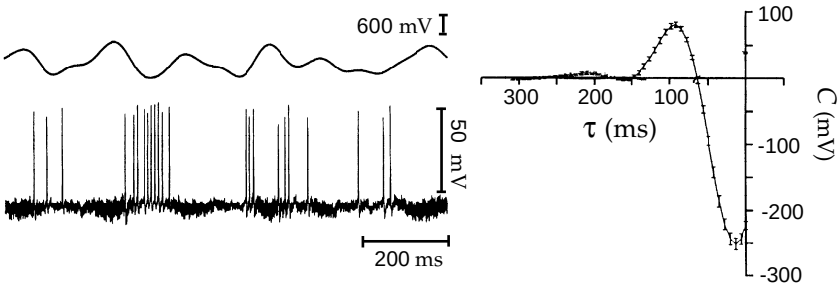

Figure 1.9 The spike-triggered average stimulus for a neuron of the electrosensory lateral-line lobe of the weakly electric fish *Eigenmannia*. The upper left trace is the potential used to generate the electric field to which this neuron is sensitive. The evoked spike train is plotted below the stimulus potential. The plot on the right is the spike-triggered average stimulus. (Adapted from Gabbiani et al., 1996.)

The spike-triggered average stimulus is widely used to study and characterize neural responses. Because $C(\tau)$ is the average value of the stimulus at a time τ before a spike, larger values of τ represent times farther in the past relative to the time of the triggering spike. For this reason, we plot spike-triggered averages with the time axis going backward compared to the normal convention. This allows the average spike-triggering stimulus to be read off from the plots in the usual left-to-right order.

Figure 1.9 shows the spike-triggered average stimulus for a neuron in the electrosensory lateral-line lobe of the weakly electric fish *Eigenmannia*. Weakly electric fish generate oscillating electric fields from an internal electric organ. Distortions in the electric field produced by nearby objects are detected by sensors spread over the skin of the fish. The lateral-line lobe acts as a relay station along the processing pathway for electrosensory signals. Fluctuating electrical potentials, such as that shown in the upper left trace of figure 1.9, elicit responses from electrosensory lateral-line lobe neurons, as seen in the lower left trace. The spike-triggered average stimulus, plotted at the right, indicates that, on average, the electric potential made a positive upswing followed by a large negative deviation prior to a spike being fired by this neuron.

The results obtained by spike-triggered averaging depend on the particular set of stimuli used during an experiment. How should this set be chosen? In chapter 2, we show that there are certain advantages to using a stimulus that is uncorrelated from one time to the next, a white-noise stimulus. A heuristic argument supporting the use of such stimuli is that in asking what makes a neuron fire, we may want to sample its responses to stimulus fluctuations at all frequencies with equal weight (i.e., equal power), and this is one of the properties of white-noise stimuli. In practice, white-noise stimuli can be generated with equal power only up to a finite frequency cutoff, but neurons respond to stimulus fluctuations only within a limited frequency range anyway. Figure 1.9 is based on such an approximate white-noise stimulus. The power in a signal as a function

of its frequency is called the power spectrum or power spectral density. White noise has a flat power spectrum.

White-Noise Stimuli

The defining characteristic of a white-noise stimulus is that its value at any one time is uncorrelated with its value at any other time. This condition can be expressed using the stimulus-stimulus correlation function, also called the stimulus autocorrelation, which is defined by analogy with equation 1.21 as

stimulus autocorrelation function Q_{ss}

$$Q_{ss}(\tau) = \frac{1}{T}\int_0^T dt\, s(t)s(t+\tau)\,. \tag{1.23}$$

Just as a correlation function provides information about the temporal relationship between two quantities, so an autocorrelation function tells us about how a quantity at one time is related to itself evaluated at another time. For white noise, the stimulus autocorrelation function is 0 in the range $-T/2 < \tau < T/2$ except when $\tau = 0$, and over this range

$$Q_{ss}(\tau) = \sigma_s^2\delta(\tau)\,. \tag{1.24}$$

The constant σ_s, which has the units of the stimulus times the square root of the unit of time, reflects the magnitude of the variability of the white noise. In appendix A, we show that equation 1.24 is equivalent to the statement that white noise has equal power at all frequencies.

No physical system can generate noise that is white to arbitrarily high frequencies. Approximations of white noise that are missing high-frequency components can be used, provided the missing frequencies are well above the sensitivity of the neuron under investigation. To approximate white noise, we consider times that are integer multiples of a basic unit of duration Δt, that is, times $t = m\Delta t$ for $m = 1, 2, \ldots, M$ where $M\Delta t = T$. The function $s(t)$ is then constructed as a discrete sequence of stimulus values. This produces a steplike stimulus waveform, like the one that appears in figure 1.8, with a constant stimulus value s_m presented during time bin m. In terms of the discrete-time values s_m, the condition that the stimulus is uncorrelated is

$$\frac{1}{M}\sum_{m=1}^{M} s_m s_{m+p} = \begin{cases} \sigma_s^2/\Delta t & \text{if} \quad p = 0 \\ 0 & \text{otherwise}\,. \end{cases} \tag{1.25}$$

The factor of $1/\Delta t$ on the right side of this equation reproduces the δ function of equation 1.24 in the limit $\Delta t \to 0$. For approximate white noise, the autocorrelation function is 0 except for a region around $\tau = 0$ with width of order Δt. Similarly, the binning of time into discrete intervals of size Δt means that the noise generated has a flat power spectrum only up to frequencies of order $1/(2\Delta t)$.

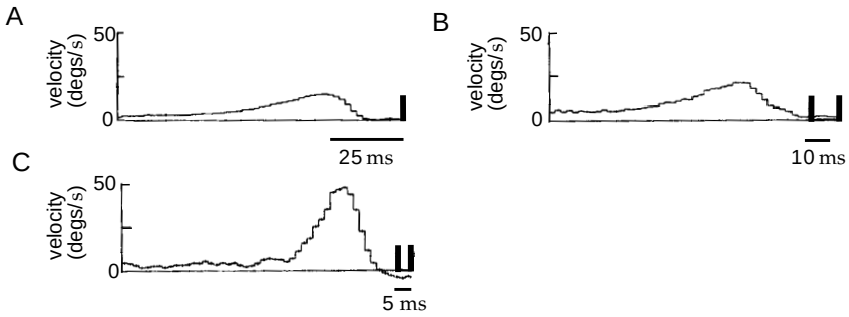

Figure 1.10 Single- and multiple-spike-triggered average stimuli for a blowfly H1 neuron responding to a moving visual image. (A) The average stimulus velocity triggered on a single spike. (B) The average stimulus velocity before two spikes with a separation of 10 ± 1 ms. (C) The average stimulus before two spikes with a separation of 5 ± 1 ms. (Data from de Ruyter van Steveninck and Bialek, 1988; figure adapted from Rieke et al., 1997.)

An approximation to white noise can be generated by choosing each s_m independently from a probability distribution with mean 0 and variance $\sigma_s^2/\Delta t$. Any reasonable probability function satisfying these two conditions can be used to generate the stimulus values within each time bin. A special class of white-noise stimuli, Gaussian white noise, results when the probability distribution used to generate the s_m values is a Gaussian function. The factor of $1/\Delta t$ in the variance indicates that the variability must be increased as the time bins get smaller. A number of other schemes for efficiently generating approximations of white-noise stimuli are discussed in the references at the end of this chapter.

Multiple-Spike-Triggered Averages and Spike-Triggered Correlations

In addition to triggering on single spikes, stimulus averages can be computed by triggering on various combinations of spikes. Figure 1.10 shows some examples of two-spike triggers. These results come from a study of the H1 movement-sensitive visual neuron of the blowfly. The H1 neuron detects the motion of visual images during flight in order to generate and guide stabilizing motor corrections. It responds to motion of the visual scene. In the experiments, the fly is held fixed while a visual image with a time-varying velocity $s(t)$ is presented. Figure 1.10A, showing the spike-triggered average stimulus, indicates that this neuron responds to positive angular velocities after a latency of about 15 ms. Figure 1.10B is the average stimulus prior to the appearance of two spikes separated by 10 ± 1 ms. In this case, the two-spike average is similar to the sum of two single-spike-triggered average stimuli displaced from one another by 10 ms. Thus, for 10 ms separations, two spikes occurring together tell us no more as a two-spike unit than they would individually. This result changes when shorter separations are considered. Figure 1.10C shows the

average stimulus triggered on two spikes separated by 5 ± 1 ms. The average stimulus triggered on a pair of spikes separated by 5 ms is not the same as the sum of the average stimuli for each spike separately.

Spike-triggered averages of other stimulus-dependent quantities can provide additional insight into neural encoding, for example, spike-triggered average autocorrelation functions. Obviously, spike-triggered averages of higher-order stimulus combinations can be considered as well.

1.4 Spike-Train Statistics

A complete description of the stochastic relationship between a stimulus and a response would require us to know the probabilities corresponding to every sequence of spikes that can be evoked by the stimulus. Spike times are continuous variables, and, as a result, the probability for a spike to occur at any precisely specified time is actually zero. To get a nonzero value, we must ask for the probability that a spike occurs within a specified interval, for example, the interval between times t and $t + \Delta t$. For small Δt, the probability of a spike falling in this interval is proportional to the size of the interval, Δt. A similar relation holds for any continuous stochastic variable z. The probability that z takes a value between z and $z + \Delta z$, for small Δz (strictly speaking, as $\Delta z \to 0$), is equal to $p[z]\Delta z$, where $p[z]$ is called a probability density.

Throughout this book, we use the notation $P[\,]$ to denote probabilities and $p[\,]$ to denote probability densities. We use the bracket notation $P[\,]$ generically for the probability of something occurring and also to denote a specific probability function. In the latter case, the notation $P(\,)$ would be more appropriate, but switching between square brackets and parentheses is confusing, so the reader will have to use the context to distinguish between these cases.

The probability of a spike sequence appearing is proportional to the probability density of spike times, $p[t_1, t_2, \ldots, t_n]$. In other words, the probability $P[t_1, t_2, \ldots, t_n]$ that a sequence of n spikes occurs with spike i falling between times t_i and $t_i + \Delta t$ for $i = 1, 2, \ldots, n$ is given in terms of this density by the relation $P[t_1, t_2, \ldots, t_n] = p[t_1, t_2, \ldots, t_n](\Delta t)^n$.

Unfortunately, the number of possible spike sequences is typically so large that determining or even roughly estimating all of their probabilities of occurrence is impossible. Instead, we must rely on some statistical model that allows us to estimate the probability of an arbitrary spike sequence occurring, given our knowledge of the responses actually recorded. The firing rate $\mathsf{r}(t)$ determines the probability of firing a single spike in a small interval around the time t, but $\mathsf{r}(t)$ is not, in general, sufficient information to predict the probabilities of spike sequences. For example, the probability of two spikes occurring together in a sequence is not necessarily equal to the product of the probabilities that they occur individually, because